# Duskmourn Report: Learned Archetypes, Bombs, and Synergies

MTGNET

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## Dataset Snapshot

This run on Duskmourn uses 216,329 unique deck builds and 1,011,949 game rows. The unsupervised clustering stage produced 10 learned archetypes. Card win rates are games-in-hand (GIH) win rates; an archetype's win rate is over all of its games. The best-performing large shells are Boros Toys | Ragged Playmate / Vicious Clown, Azorius | Tunnel Surveyor / Glimmerburst, Simic | Oblivious Bookworm / Growing Dread, which already tells us that the format is not a single monolithic color table: it separates into distinct deck families with their own signposts, core cards, and standalone bombs.

## How to Read This Report

- **Coreness** measures how central a card is to an archetype: played in many of its decks, characteristic of it, and connected to the rest of the deck. Cards are grouped into Core, Support and Periphery rings. A core card may carry a tag: **signature** (in bold: played in at least 15% of this archetype's decks and in at most one other archetype at that rate) or *staple* (at least 25% of this archetype's decks and at least 15% in three or more archetypes); untagged core cards are neither.
- **Play rate** is the share of the archetype's decks that play at least one copy.
- **Bombs** are cards that are strong on their own. A bomb's **lift** compares, within the decks that play it, the win rate when the card is in hand with the win rate when it stayed in the library, relative to a typical card of the archetype, and is shrunk across archetypes; it is shown in percentage points (pp). Bombs are listed whatever ring they sit in: core and support bombs keep their ring on the site, and only bombs that would otherwise be peripheral go to the bomb island beside the rings.
- A **synergy pair** wins more often when both cards are drawn than their separate effects predict (games-in-hand, adjusted for game length and play/draw). The effect $\lambda$ is an interaction log-odds, shown as a percentage change in the odds of winning with its 95% confidence interval. Pairs are tested with Benjamini–Hochberg false-discovery control: *verified* means $q \leq 0.10$, *tentative* means $q \leq 0.25$. Few pairs pass; an absent pair means "not detectable with this data", not "no synergy".
- **Payoff** and **engine** roles are decided by synergy arrows only: the synergy pairs (verified or tentative) whose direction is significant. In such a pair the arrow points from the payoff, the

card that wins more when drawn without its partner, to the engine it rewards; a card whose arrows mostly point out of it is a payoff, one whose arrows mostly point into it an engine. The arrow reflects each card's standalone strength within the pair, not a direction of the synergy itself. Played-together links carry the same arrow on the site as context only; they do not set roles.

# 1 Discovered Archetypes

| Archetype | Decks | Games | WR | Top signposts |
|---|---|---|---|---|
| Boros Toys \| Ragged Playmate / Vicious Clown | 29,938 | 147,650 | 56.77% | Ragged Playmate, Vicious Clown, Midnight Mayhem, Turn Inside Out, Arabella, Abandoned Doll |
| Azorius \| Tunnel Surveyor / Glimmerburst | 33,694 | 163,696 | 56.14% | Tunnel Surveyor, Glimmerburst, Grand Entryway // Elegant Rotunda, Gremlin Tamer, Unable to Scream |
| Simic \| Oblivious Bookworm / Growing Dread | 29,996 | 139,936 | 55.14% | Oblivious Bookworm, Growing Dread, Cryptid Inspector, Manifest Dread, Paranormal Analyst |
| Gruul \| Beastie Beatdown / Wildfire Wickerfolk | 27,260 | 130,402 | 54.97% | Beastie Beatdown, Wildfire Wickerfolk, Spineseeker Centipede, Wary Watchdog, Scorching Dragonfire |
| Rakdos \| Disturbing Mirth / Final Vengeance | 19,451 | 89,951 | 54.60% | Disturbing Mirth, Final Vengeance, Sawblade Skinripper, Innocuous Rat, Fear of Lost Teeth |
| Golgari \| Broodspinner / Drag to the Roots | 22,174 | 102,257 | 53.95% | Broodspinner, Drag to the Roots, Say Its Name, Spineseeker Centipede, Murder |
| Selesnya Survivors \| Shrewd Storyteller / Baseball Bat | 11,863 | 55,207 | 52.13% | Shrewd Storyteller, Baseball Bat, Acrobatic Cheerleader, Hardened Escort, Flesh Burrower |
| Orzhov \| Emerge from the Cocoon / Shroudstomper | 15,217 | 66,755 | 51.50% | Emerge from the Cocoon, Shroudstomper, Rite of the Moth, Fanatic of the Harrowing, Spectral Snatcher |
| Izzet \| Smoky Lounge // Misty Salon / Rampaging Soulrager | 14,682 | 64,564 | 51.13% | Smoky Lounge // Misty Salon, Rampaging Soulrager, Intruding Soulrager, Glassworks // Shattered Yard, Meat Locker // Drowned Diner |

| Archetype | Decks | Games | WR | Top signposts |
|---|---|---|---|---|
| Dimir \| Fear of Infinity / Skullsnap Nuisance | 12,054 | 51,531 | 51.06% | Fear of Infinity, Skullsnap Nuisance, Balemurk Leech, Murder, Unable to Scream |

## 2 Core of Each Archetype

The core ring of each archetype, most central first, with the share of the archetype's decks that play each card and its tag where it has one: **signature** (in bold) or staple. The second line lists the archetype's bombs, whatever ring they sit in: cards that are strong on their own, with their lift (in-hand vs in-library win rate within the decks that play them, relative to a typical card of the archetype, shrunk across archetypes) and their games in hand.

| Archetype | Core cards (play rate) |
|---|---|
| Azorius<br>33,694 decks | Grand Entryway // Elegant Rotunda (55%, staple), **Gremlin Tamer** (46%, signature), **Inquisitive Glimmer** (43%, signature), Glimmerburst (58%, staple), Tunnel Surveyor (59%, staple), Unable to Scream (55%, staple), Stalked Researcher (43%, staple), Fear of Immobility (43%, staple), Fear of Surveillance (49%, staple), Meat Locker // Drowned Diner (52%, staple)<br>*Bombs:* Overlord of the Mistmoors (+14.5 pp, 3,640 games); Ghostly Dancers (+12.1 pp, 10,337 games); Dollmaker's Shop // Porcelain Gallery (+9.7 pp, 3,721 games); Valgavoth's Onslaught (+8.1 pp, 1,800 games) |
| Simic<br>29,996 decks | **Oblivious Bookworm** (55%, signature), Cryptid Inspector (47%, staple), Manifest Dread (57%, staple), **Paranormal Analyst** (39%, signature), **Growing Dread** (49%, signature), Twist Reality (38%, staple), Bashful Beastie (50%, staple), Monstrous Emergence (49%, staple), Moldering Gym // Weight Room (43%, staple), Unable to Scream (48%, staple)<br>*Bombs:* Valgavoth's Onslaught (+13.2 pp, 7,634 games); The Swarmweaver (+9.9 pp, 4,488 games); Ghostly Dancers (+9.5 pp, 815 games); The Rollercrusher Ride (+8.1 pp, 1,193 games) |
| Boros Toys<br>29,938 decks | **Vicious Clown** (52%, signature), Turn Inside Out (57%, staple), **Midnight Mayhem** (49%, signature), Ragged Playmate (60%, staple), **Arabella, Abandoned Doll** (51%, signature), Clockwork Percussionist (59%, staple), Most Valuable Slayer (57%, staple), Jump Scare (42%, staple), Acrobatic Cheerleader (44%, staple), Unsettling Twins (46%, staple)<br>*Bombs:* Overlord of the Mistmoors (+14.5 pp, 2,487 games); Valgavoth's Onslaught (+8.6 pp, 829 games); Dollmaker's Shop // Porcelain Gallery (+8.2 pp, 3,038 games); Ghostly Dancers (+7.6 pp, 4,329 games) |
| Gruul<br>27,260 decks | **Wildfire Wickerfolk** (52%, signature), Spineseeker Centipede (54%, staple), Wary Watchdog (52%, staple), Hand That Feeds (47%, staple), Most Valuable Slayer (49%, staple), Say Its Name (41%, staple), Turn Inside Out (44%, staple), Clockwork Percussionist (46%, staple), Monstrous Emergence (46%, staple), Scorching Dragonfire (56%, staple)<br>*Bombs:* Valgavoth's Onslaught (+13.5 pp, 5,988 games); Unholy Annex // Ritual Chamber (+10.2 pp, 821 games); The Swarmweaver (+9.2 pp, 3,727 games); Overlord of the Boilerbilges (+8.7 pp, 3,116 games) |

| Archetype | Core cards (play rate) |
|---|---|
| Golgari<br>22,174 decks | Say Its Name (50%, staple), Murder (45%, staple), **Broodspinner** (51%, signature), Spineseeker Centipede (55%, staple), Wary Watchdog (50%, staple), Monstrous Emergence (47%, staple), Winter's Intervention (39%, staple), Innocuous Rat (38%, staple), **Drag to the Roots** (47%, signature), Spectral Snatcher (36%, staple)<br>*Bombs:* Valgavoth's Onslaught (+13.4 pp, 4,755 games); Overlord of the Mistmoors (+13.3 pp, 542 games); The Swarmweaver (+12.2 pp, 10,568 games); Ghostly Dancers (+11.7 pp, 433 games) |
| Rakdos<br>19,451 decks | Final Vengeance (66%, staple), **Disturbing Mirth** (59%, signature), Innocuous Rat (59%, staple), Fear of Lost Teeth (51%, staple), Clockwork Percussionist (60%, staple), **Sawblade Skinripper** (54%, signature), **Vengeful Possession** (37%, signature), **Popular Egotist** (36%, signature), Hand That Feeds (42%, staple), **Boilerbilges Ripper** (29%, signature)<br>*Bombs:* Unholy Annex // Ritual Chamber (+11.1 pp, 5,153 games); The Swarmweaver (+9.4 pp, 1,585 games); Valgavoth's Onslaught (+8.1 pp, 1,129 games); Unstoppable Slasher (+7.9 pp, 5,284 games) |
| Orzhov<br>15,217 decks | **Emerge from the Cocoon** (57%, signature), **Shroudstomper** (53%, signature), **Rite of the Moth** (50%, signature), Spectral Snatcher (52%, staple), Fanatic of the Harrowing (46%, staple), **Possessed Goat** (34%, signature), Shepherding Spirits (40%, staple), Live or Die (33%, staple), **Valgavoth's Faithful** (33%, signature), Vile Mutilator (32%, staple)<br>*Bombs:* Overlord of the Mistmoors (+13.8 pp, 1,848 games); Unholy Annex // Ritual Chamber (+12.0 pp, 4,030 games); Valgavoth's Onslaught (+12.0 pp, 800 games); The Swarmweaver (+9.2 pp, 1,117 games) |
| Izzet<br>14,682 decks | **Smoky Lounge // Misty Salon** (59%, signature), **Rampaging Soulrager** (44%, signature), Glassworks // Shattered Yard (63%, staple), **Intruding Soulrager** (45%, signature), Creeping Peeper (43%, staple), Meat Locker // Drowned Diner (59%, staple), Underwater Tunnel // Slimy Aquarium (52%, staple), Stalked Researcher (44%, staple), Ticket Booth // Tunnel of Hate (47%, staple), **Roaring Furnace // Steaming Sauna** (30%, signature)<br>*Bombs:* Roaring Furnace // Steaming Sauna (+9.9 pp, 9,160 games); Unholy Annex // Ritual Chamber (+9.3 pp, 447 games); Overlord of the Boilerbilges (+8.6 pp, 1,497 games); Valgavoth's Onslaught (+8.4 pp, 909 games) |
| Dimir<br>12,054 decks | Balemurk Leech (35%, staple), Murder (44%, staple), Appendage Amalgam (31%, staple), Meat Locker // Drowned Diner (50%, staple), **Skullsnap Nuisance** (51%, signature), Final Vengeance (39%, staple), Fear of Lost Teeth (33%, staple), Glimmerburst (49%, staple), Winter's Intervention (40%, staple), **Dashing Bloodsucker** (27%, signature)<br>*Bombs:* Unholy Annex // Ritual Chamber (+14.0 pp, 3,693 games); Overlord of the Mistmoors (+13.9 pp, 277 games); Valgavoth's Onslaught (+12.1 pp, 654 games); Ghostly Dancers (+10.4 pp, 455 games) |

| Archetype | Core cards (play rate) |
|---|---|
| Selesnya Survivors<br>11,863 decks | Acrobatic Cheerleader (47%, staple), **Shrewd Storyteller** (49%, signature), **Baseball Bat** (47%, signature), **Hardened Escort** (37%, signature), **Horrid Vigor** (30%, signature), **Orphans of the Wheat** (29%, signature), Friendly Ghost (35%, staple), Jump Scare (33%, staple), **Cautious Survivor** (28%, signature), Flesh Burrower (47%, staple)<br>*Bombs:* Overlord of the Mistmoors (+14.7 pp, 1,162 games); Unholy Annex // Ritual Chamber (+11.1 pp, 234 games); Valgavoth's Onslaught (+10.9 pp, 2,184 games); Unidentified Hovership (+9.0 pp, 2,851 games) |

# 3 What Defines the Best Shells

### Boros Toys | Ragged Playmate / Vicious Clown

This cluster contains 29,938 decks across 147,650 games and posts a baseline win rate of 56.77%. Its core (the cards most central to it) is Vicious Clown (52% of decks), Turn Inside Out (57% of decks), Midnight Mayhem (49% of decks), Ragged Playmate (60% of decks), Arabella, Abandoned Doll (51% of decks). Its defining signposts are Ragged Playmate, Vicious Clown, Midnight Mayhem, Turn Inside Out, Arabella, Abandoned Doll, Clockwork Percussionist.

Bombs that win on their own here: Overlord of the Mistmoors (+14.5 pp), Valgavoth's Onslaught (+8.6 pp), Dollmaker's Shop // Porcelain Gallery (+8.2 pp). Out of 496 card pairs tested, 4 are verified synergies (BH q $\leq$ 0.10) and 13 tentative (q $\leq$ 0.25). The strongest verified pair is Midnight Mayhem + Painter's Studio // Defaced Gallery: +42% odds of winning when both are drawn (95% CI [+31, +54]%, 4,217 games). Among its synergy pairs with a significant direction, the payoffs (cards that win more when drawn than their partners) are Dollmaker's Shop // Porcelain Gallery, Midnight Mayhem and the engines they reward are Friendly Ghost.

### Azorius | Tunnel Surveyor / Glimmerburst

This cluster contains 33,694 decks across 163,696 games and posts a baseline win rate of 56.14%. Its core (the cards most central to it) is Grand Entryway // Elegant Rotunda (55% of decks), Gremlin Tamer (46% of decks), Inquisitive Glimmer (43% of decks), Glimmerburst (58% of decks), Tunnel Surveyor (59% of decks). Its defining signposts are Tunnel Surveyor, Glimmerburst, Grand Entryway // Elegant Rotunda, Gremlin Tamer, Unable to Scream, Inquisitive Glimmer.

Bombs that win on their own here: Overlord of the Mistmoors (+14.5 pp), Ghostly Dancers (+12.1 pp), Dollmaker's Shop // Porcelain Gallery (+9.7 pp). Out of 496 card pairs tested, 20 are verified synergies (BH q $\leq$ 0.10) and 27 tentative (q $\leq$ 0.25). The strongest verified pair is Grand Entryway // Elegant Rotunda + Optimistic Scavenger: +20% odds of winning when both are drawn (95% CI [+13, +28]%, 8,608 games). Among its synergy pairs with a significant direction, the payoffs (cards that win more when drawn than their partners) are Entity Tracker, Gremlin Tamer, Optimistic Scavenger and the engines they reward are Grand Entryway // Elegant Rotunda, Creeping Peeper, Fear of Surveillance.

### Simic | Oblivious Bookworm / Growing Dread

This cluster contains 29,996 decks across 139,936 games and posts a baseline win rate of 55.14%. Its core (the cards most central to it) is `Oblivious Bookworm` (55% of decks), `Cryptid Inspector` (47% of decks), `Manifest Dread` (57% of decks), `Paranormal Analyst` (39% of decks), `Growing Dread` (49% of decks). Its defining signposts are `Oblivious Bookworm`, `Growing Dread`, `Cryptid Inspector`, `Manifest Dread`, `Paranormal Analyst`, `Bashful Beastie`.

Bombs that win on their own here: `Valgavoth's Onslaught` (+13.2 pp), `The Swarmweaver` (+9.9 pp), `Ghostly Dancers` (+9.5 pp). Out of 496 card pairs tested, 14 are verified synergies (BH $q \leq 0.10$) and 12 tentative ($q \leq 0.25$). The strongest verified pair is `Paranormal Analyst` + `Underwater Tunnel // Slimy Aquarium`: +19% odds of winning when both are drawn (95% CI [+12, +27]%, 6,180 games). Among its synergy pairs with a significant direction, the payoffs (cards that win more when drawn than their partners) are `Oblivious Bookworm`, `Under the Skin`, `Growing Dread` and the engines they reward are `Monstrous Emergence`, `Paranormal Analyst`, `Cryptid Inspector`.

### Gruul | Beastie Beatdown / Wildfire Wickerfolk

This cluster contains 27,260 decks across 130,402 games and posts a baseline win rate of 54.97%. Its core (the cards most central to it) is `Wildfire Wickerfolk` (52% of decks), `Spineseeker Centipede` (54% of decks), `Wary Watchdog` (52% of decks), `Hand That Feeds` (47% of decks), `Most Valuable Slayer` (49% of decks). Its defining signposts are `Beastie Beatdown`, `Wildfire Wickerfolk`, `Spineseeker Centipede`, `Wary Watchdog`, `Scorching Dragonfire`, `Hand That Feeds`.

Bombs that win on their own here: `Valgavoth's Onslaught` (+13.5 pp), `Unholy Annex // Ritual Chamber` (+10.2 pp), `The Swarmweaver` (+9.2 pp). Out of 496 card pairs tested, 5 are verified synergies (BH $q \leq 0.10$) and 6 tentative ($q \leq 0.25$). The strongest verified pair is `Clockwork Percussionist` + `Turn Inside Out`: +17% odds of winning when both are drawn (95% CI [+10, +25]%, 7,287 games). Among its synergy pairs with a significant direction, the payoffs (cards that win more when drawn than their partners) are `Spineseeker Centipede` and the engines they reward are `Most Valuable Slayer`.

## 4 Bombs: Cards That Are Good Alone

Cards flagged as bombs in at least one archetype, whatever ring they sit in on the site, each shown in the archetype where its lift is largest. *Lift*: within the decks that play the card, the win rate when it is in hand vs when it stayed in the library, relative to a typical card of the archetype, shrunk across archetypes. *Raw lift*: the card's games-in-hand win rate minus the archetype baseline, not shrunk.

| Card | Archetype | Lift | Raw lift | GIH WR | Games |
|---|---|---|---|---|---|
| `Overlord of the Mistmoors` | Selesnya Survivors | +14.7 pp | +15.9 pp | 68.07% | 1,162 |
| `Unholy Annex // Ritual Chamber` | Dimir | +14.0 pp | +14.8 pp | 65.83% | 3,693 |
| `Valgavoth's Onslaught` | Gruul | +13.5 pp | +13.2 pp | 68.17% | 5,988 |
| `The Swarmweaver` | Golgari | +12.2 pp | +12.2 pp | 66.16% | 10,568 |
| `Ghostly Dancers` | Azorius | +12.1 pp | +11.1 pp | 67.23% | 10,337 |

| Card | Archetype | Lift | Raw lift | GIH WR | Games |
|---|---|---|---|---|---|
| Roaring Furnace // Steaming Sauna | Izzet | +9.9 pp | +10.5 pp | 61.59% | 9,160 |
| Dollmaker's Shop // Porcelain Gallery | Azorius | +9.7 pp | +10.3 pp | 66.43% | 3,721 |
| Unidentified Hovership | Selesnya Survivors | +9.0 pp | +8.2 pp | 60.33% | 2,851 |
| Dissection Tools | Golgari | +9.0 pp | +9.6 pp | 63.59% | 2,771 |
| Overlord of the Balemurk | Orzhov | +8.9 pp | +12.6 pp | 64.13% | 1,957 |
| Overlord of the Boilerbilges | Gruul | +8.7 pp | +11.4 pp | 66.34% | 3,116 |
| Overlord of the Floodpits | Dimir | +8.4 pp | +12.6 pp | 63.70% | 1,595 |
| The Rollercrusher Ride | Simic | +8.1 pp | +9.0 pp | 64.12% | 1,193 |
| Zimone, All-Questioning | Simic | +8.1 pp | +8.6 pp | 63.75% | 13,535 |

# 5 Build-Arounds and Shell Dependence

| Card | Best cluster | Shell dependence | Support delta | Supported games |
|---|---|---|---|---|
| Turn Inside Out | Boros Toys \| Ragged Playmate / Vicious Clown | 10.0 / 10 | 20.03 pp | 39299 |
| Hand That Feeds | Rakdos \| Disturbing Mirth / Final Vengeance | 10.0 / 10 | 31.14 pp | 18943 |
| Grand Entryway // Elegant Rotunda | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 21.03 pp | 47525 |
| Stalked Researcher | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 19.56 pp | 34749 |
| Moldering Gym // Weight Room | Simic \| Oblivious Bookworm / Growing Dread | 10.0 / 10 | 20.77 pp | 29672 |
| Cryptid Inspector | Simic \| Oblivious Bookworm / Growing Dread | 10.0 / 10 | 19.94 pp | 34885 |
| Appendage Amalgam | Dimir \| Fear of Infinity / Skullsnap Nuisance | 10.0 / 10 | 23.83 pp | 7721 |
| Erratic Apparition | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 19.74 pp | 25498 |
| Doomsday Excruciator | Orzhov \| Emerge from the Cocoon / Shroudstomper | 9.9 / 10 | 29.93 pp | 1918 |
| Derelict Attic // Widow's Walk | Rakdos \| Disturbing Mirth / Final Vengeance | 9.9 / 10 | 32.04 pp | 8466 |
| Monstrous Emergence | Gruul \| Beastie Beatdown / Wildfire Wickerfolk | 9.8 / 10 | 19.13 pp | 27388 |

| Card | Best cluster | Shell dependence | Support delta | Supported games |
|---|---|---|---|---|
| Boilerbilges Ripper | Rakdos \| Disturbing Mirth / Final Vengeance | 9.7 / 10 | 25.57 pp | 11613 |

## 6 Dependency Cards

| Card | Best cluster | Dependency | Standalone lift | Games |
|---|---|---|---|---|
| Central Elevator // Promising Stairs | Izzet \| Smoky Lounge // Misty Salon / Rampaging Soulrager | 0.474 | 1.65 pp | 4750 |
| Moldering Gym // Weight Room | Simic \| Oblivious Bookworm / Growing Dread | 0.456 | -0.55 pp | 29790 |
| Ghostly Dancers | Simic \| Oblivious Bookworm / Growing Dread | 0.456 | 8.79 pp | 815 |
| Doomsday Excruciator | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.410 | -7.40 pp | 1925 |
| Appendage Amalgam | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.354 | -0.17 pp | 11382 |
| Valgavoth, Terror Eater | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.353 | 4.67 pp | 2656 |
| Painter's Studio // Defaced Gallery | Boros Toys \| Ragged Playmate / Vicious Clown | 0.352 | 3.52 pp | 18515 |
| Enduring Curiosity | Dimir \| Fear of Infinity / Skullsnap Nuisance | 0.336 | 9.64 pp | 3112 |
| Fear of Lost Teeth | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.331 | 1.54 pp | 10602 |
| Unholy Annex // Ritual Chamber | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.298 | 13.16 pp | 4030 |
| Possessed Goat | Orzhov \| Emerge from the Cocoon / Shroudstomper | 0.264 | 1.58 pp | 11527 |
| Drag to the Roots | Gruul \| Beastie Beatdown / Wildfire Wickerfolk | 0.264 | 0.23 pp | 3627 |

# 7 Best Draft Priorities

| Card | Best cluster | Pick score | Draft win lift | Rarity |
|---|---|---|---|---|
| The Swarmweaver | Golgari \| Broodspinner / Drag to the Roots | 10.0 / 10 | 3.89 pp | rare |
| Broodspinner | Golgari \| Broodspinner / Drag to the Roots | 10.0 / 10 | 1.74 pp | uncommon |
| Oblivious Bookworm | Simic \| Oblivious Bookworm / Growing Dread | 10.0 / 10 | 1.31 pp | uncommon |
| Roaring Furnace // Steaming Sauna | Izzet \| Smoky Lounge // Misty Salon / Rampaging Soulrager | 10.0 / 10 | 2.25 pp | rare |
| Disturbing Mirth | Rakdos \| Disturbing Mirth / Final Vengeance | 10.0 / 10 | 1.31 pp | uncommon |
| Midnight Mayhem | Boros Toys \| Ragged Playmate / Vicious Clown | 10.0 / 10 | 1.33 pp | uncommon |
| Ghostly Dancers | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 3.32 pp | rare |
| Zimone, All-Questioning | Simic \| Oblivious Bookworm / Growing Dread | 10.0 / 10 | 2.56 pp | rare |
| Wildfire Wickerfolk | Gruul \| Beastie Beatdown / Wildfire Wickerfolk | 10.0 / 10 | 1.44 pp | uncommon |
| Optimistic Scavenger | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 1.81 pp | uncommon |
| Valgavoth's Onslaught | Gruul \| Beastie Beatdown / Wildfire Wickerfolk | 10.0 / 10 | 4.49 pp | rare |
| Gremlin Tamer | Azorius \| Tunnel Surveyor / Glimmerburst | 10.0 / 10 | 1.14 pp | uncommon |

# 8 Verified Synergy Pairs

Every pair below passed a pair-interaction test on games-in-hand data, stratified by game length and play/draw, with Benjamini–Hochberg false-discovery control across all pairs tested in its archetype. $\lambda$ is the interaction log-odds, shown as the change in the odds of winning when both cards are drawn, with its 95% confidence interval. Across all archetypes, 92 pairs are verified ($q \leq 0.10$) and 118 tentative ($q \leq 0.25$, marked †).

| Pair | | Archetype | Odds | 95% CI | $q$ | Pair WR | Games |
|---|---|---|---|---|---|---|---|
| Midnight Mayhem<br>Painter's Studio // Defaced Gallery | + | Boros Toys | +42% | [+31, +54]% | <0.001 | 70.78% | 4,217 |

| Pair | Archetype | Odds | 95% CI | $q$ | Pair WR | Games |
|---|---|---|---|---|---|---|
| Disturbing Mirth + Final Vengeance | Rakdos | +27% | [+20, +36]% | <0.001 | 65.85% | 10,731 |
| Central Elevator // Promising Stairs + Smoky Lounge // Misty Salon | Izzet | +61% | [+41, +83]% | <0.001 | 61.17% | 1,813 |
| Grand Entryway // Elegant Rotunda + Optimistic Scavenger | Azorius | +20% | [+13, +28]% | <0.001 | 65.67% | 8,608 |
| Final Vengeance + Innocuous Rat | Rakdos | +18% | [+11, +25]% | <0.001 | 62.61% | 10,919 |
| Paranormal Analyst + Underwater Tunnel // Slimy Aquarium | Simic | +19% | [+12, +27]% | <0.001 | 59.48% | 6,180 |
| Entity Tracker + Grand Entryway // Elegant Rotunda | Azorius | +26% | [+15, +39]% | <0.001 | 68.14% | 3,220 |
| Grand Entryway // Elegant Rotunda + Gremlin Tamer | Azorius | +14% | [+8, +20]% | <0.001 | 63.16% | 11,525 |
| Emerge from the Cocoon + Fanatic of the Harrowing | Orzhov | +23% | [+13, +33]% | <0.001 | 60.69% | 5,754 |
| Final Vengeance + Vengeful Possession | Rakdos | +21% | [+12, +30]% | <0.001 | 64.20% | 6,271 |
| Entity Tracker + Unable to Scream | Azorius | +25% | [+14, +37]% | <0.001 | 68.02% | 3,396 |
| Doomsday Excruciator + Fanatic of the Harrowing | Orzhov | +63% | [+33, +98]% | <0.001 | 55.95% | 672 |
| Possessed Goat + Valgavoth, Terror Eater | Orzhov | +60% | [+32, +94]% | <0.001 | 65.52% | 638 |
| Smoky Lounge // Misty Salon + Underwater Tunnel // Slimy Aquarium | Izzet | +20% | [+12, +30]% | <0.001 | 56.96% | 6,283 |
| Clockwork Percussionist + Turn Inside Out | Gruul | +17% | [+10, +25]% | <0.001 | 59.85% | 7,287 |
| Erratic Apparition + Unable to Scream | Azorius | +14% | [+8, +22]% | 0.001 | 58.60% | 7,558 |
| Derelict Attic // Widow's Walk + Final Vengeance | Rakdos | +24% | [+13, +36]% | 0.001 | 62.62% | 3,365 |
| Inquisitive Glimmer + Meat Locker // Drowned Diner | Azorius | +13% | [+6, +19]% | 0.002 | 61.02% | 9,431 |
| Emerge from the Cocoon + Valgavoth, Terror Eater | Orzhov | +42% | [+21, +68]% | 0.003 | 63.67% | 1,079 |
| Disturbing Mirth + Fear of Lost Teeth | Rakdos | +15% | [+7, +22]% | 0.003 | 62.57% | 7,740 |
| Gremlin Tamer + Unable to Scream | Azorius | +11% | [+6, +17]% | 0.004 | 62.60% | 11,306 |
| Entity Tracker + Fear of Surveillance | Azorius | +21% | [+10, +33]% | 0.004 | 67.07% | 2,909 |
| Stalked Researcher + Trapped in the Screen | Azorius | +12% | [+6, +18]% | 0.004 | 59.09% | 9,227 |

| Pair | Archetype | Odds | 95% CI | $q$ | Pair WR | Games |
|---|---|---|---|---|---|---|
| Ethereal Armor + Fear of Surveillance | Azorius | +14% | [+7, +23]% | 0.004 | 62.02% | 5,598 |

## 9 Anti-Synergies Worth Respecting

Pairs whose interaction is significantly negative ($q \leq 0.10$): they win less often when drawn together than their separate effects predict.

| Pair | Archetype | Odds | 95% CI | Pair WR | Games |
|---|---|---|---|---|---|
| Ghostly Dancers + Overlord of the Mistmoors | Azorius | -42% | [-61, -14]% | 70.45% | 132 |
| Underwater Tunnel // Slimy Aquarium + Valgavoth's Onslaught | Izzet | -40% | [-56, -18]% | 52.19% | 228 |
| Dollmaker's Shop // Porcelain Gallery + Painter's Studio // Defaced Gallery | Boros Toys | -40% | [-53, -23]% | 57.32% | 314 |
| Overlord of the Balemurk + Unsettling Twins | Orzhov | -39% | [-52, -22]% | 59.66% | 352 |
| Nashi, Searcher in the Dark + Unholy Annex // Ritual Chamber | Dimir | -33% | [-47, -16]% | 60.71% | 364 |
| Defiant Survivor + Rip, Spawn Hunter | Selesnya Survivors | -33% | [-44, -19]% | 47.63% | 569 |
| Moldering Gym // Weight Room + Ticket Booth // Tunnel of Hate | Izzet | -30% | [-44, -13]% | 48.19% | 415 |
| Ghostly Dancers + Live or Die | Orzhov | -28% | [-42, -11]% | 58.85% | 452 |
| Under the Skin + Valgavoth's Onslaught | Gruul | -28% | [-40, -14]% | 66.88% | 619 |
| Most Valuable Slayer + The Swarmweaver | Gruul | -26% | [-38, -12]% | 57.32% | 656 |

## 10 Bottom-Line Reading

This report treats the format as a deck ecosystem rather than a flat card ranking. Each archetype is described first by its core: the cards that are played in most of its decks, characteristic of it and connected to the rest of the deck. Bombs are listed separately because they win on their own and are worth taking (and often splashing) whatever the shell. Synergy claims are deliberately conservative: only pairs that survive a false-discovery-controlled interaction test are reported, with their effect size and confidence interval, so the short pair list is a feature of honest statistics rather than a lack of interplay between the cards.